#Probability, #Statistics

Notebook

Day 1

Probability and Statistics.

Modeling the world: uncertain.

Modeling uncertainty ... with Data.

Examples:

* Gas molecules → Temperature, density, etc.
* Kalman filter → stochastic Diff. EQ.
* weather (chaotic)
* Human behavior.

Probability: → assume known prob. distribution
→ samples unknown

Statistics ⇒ Data (Samples) known.
... prob. distribution unknown.

Contents:

(I) Intro to probability.

- Examples / Intuition.
- Counting sets.

(II) Random Variable X & functions / Distributions.

$$X \sim p(X = \_ \,/\, \theta)$$

data; θ → parameters.

- Bernoulli
- Binomial
- exponential
- Normal
- poisson
- ...

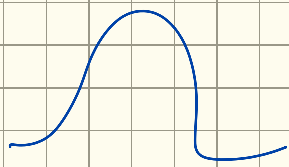
- Sometimes distribution is unknown (ML)

(III) Expected value, variance, SD, median.

(IV) Central Limit Theorem

X_i, $1, 2, \ldots, n$

$$\bar{X} = \frac{1}{n}\sum X_i \sim \text{normal}(\mu, \sigma^2)$$



$$\not\Downarrow$$

Statistics:

$$p(\theta \mid X)$$

params ↲ ↳ data.

- Hypothesis testing.
- Survey sampling.

fitting, estimating ... ML

probability:

Prob. of an event 'A'

$$P(A) = \frac{\text{\# of ways A can happen.}}{\text{\# of things that can happen.}}$$

Counting method:

Ex: Flip a coin twice. $\rightarrow \{hh, ht, th, tt\} = \Omega$

1. A = at least one heads.

hh, ht, th

$A = \{hh, ht, th\}$.

$\boxed{P(A) = \frac{3}{4}} = 75\%$

2. B = no heads at all

$B = \{tt\}$.

$P(B) = \frac{1}{4} = 25\%$.

Ex: I roll two dice (6-sided dice)

I.) prob. (at least one die is a 5)? A

$\Omega = \{11, 12, 13, 14, 15, \ldots 52, 53, \ldots, 66\}$

$A = \{15, 25, 35, 45, 51, 52, 53, 54, 55, 56, 65\}$

$\boxed{P(A) = \frac{11}{36}}$

OR

$P(A) = P(D_1 = 5) + P(D_1 \neq 5) \cdot P(D_2 = 5)$

$= \frac{1}{6} + \frac{5}{6} \cdot \frac{1}{6} = \boxed{\frac{11}{36}}.$

What do we mean by random?

$\Rightarrow$ The idea that the outcome of an action is not deterministic or is very hard to model deterministically.

Being clever: counting easily:

- Combinatorics : counting combinations.

Ex:
# of sequences of 10 coin flips:
$\hookrightarrow$ order matters. ($HT \neq TH$)

$\Rightarrow 2 \times 2 \times 2 \times \cdots \times 2$

$\Rightarrow \boxed{2^{10}}$ (with replacement)

Deck of 52 cards how many 5 card runs can I deal off top of deck.

$(n! = n \times (n-1) \times \cdots \times 3 \times 2 \times 1)$

$$\underline{52} \times \underline{51} \times \underline{50} \times \underline{49} \times \underline{48} = \frac{52!}{47!}$$

(without Replacement)
(order matters)

n choices
sampling r

$\# = n^r$ (with replacement)

$\# = \dfrac{n!}{(n-r)!}$ (without replacement)

If order doesn't matter: $\# = \dfrac{n!}{(n-r)!\, r!} = \binom{n}{r}$ $= {}_nC_r$

(n choose r)

Ex: Licence Plates:

_ _ _ _   _ _

Alphabets   numbers.

with replacement $= 26^4 \times 10^2$

without replacement (all unique). $= \frac{26!}{(26-4)!} \times \frac{10!}{(10-2)!}$

Some Questions:

1) 2 fair coins flipped, $P(\overbrace{\text{at least one head}}^{A})$ computed via the Complement rule?

$$\Omega = \{HH, HT, TH, TT\}$$

$$A = \{HH, HT, TH\}$$

$$P(A) = \frac{3}{4}$$

In large sets $P(A) = P(\text{exactly 1 head}) + P(\text{exactly 2}) + \ldots$

OR

using complement rule, (easier because opposite case appearance only once in the whole sample space).

$$P(A) = 1 - P(\text{not any head})$$

$$= 1 - \frac{1}{4}$$

$$= \boxed{\frac{3}{4}}$$

$$P(A) = 1 - P(\bar{A})$$

$$= 1 - \left(\frac{1}{2}\right)^r$$

2) $\binom{n}{r} = \frac{n!}{(n-r)!\,r!}$, order doesn't matter, without replacement.

$\frac{n!}{(n-r)!}$ gives for n items in which order matters & without replacement.

$r!$ is the result that they can arrange themselves.

3)

$\Rightarrow$ LEVEL

Distinct ways

5 __ ? __ ? __ ? __ ? __ ??

Think!!

$$\frac{5!}{0! \times 2! \times 2!} = \frac{5 \times \cancel{4}^{2} \times 3}{\cancel{2}}$$

4) Fair coin lands heads 10 times in a row. Does that change the probability the 11^{th} flip is heads?

$\Rightarrow$ If fair coin, flipping a coin is random and 11^{th} flip should be independent of the 10^{th} flip. Hence, still $p = \frac{1}{2}$.

People $\rightarrow$ emotional creatures $\rightarrow$ gambler's fallacy.

$\downarrow$

think same would appear again.

5) 4 of 10 books on a shelf?

$\Rightarrow$

$\underline{10} \times \underline{9} \times \underline{8} \times \underline{7}$

6) 4-digit PIN, repeats allowed: (0-9)

$\underline{10}\ \underline{10}\ \underline{10}\ \underline{10} \Rightarrow 10^4$

7) $\binom{8}{3} = \frac{8!}{5!\,3!} = \frac{8 \times 7 \times \cancel{6}}{\cancel{6}} = 56$

8) __ __ __ __ __

52 cards

5-card hand..?

$\frac{52!}{(52-5)!\,5!}$

9) __ __ __   __ __ __

Letters   digits   (repetition allowed)

$26^3 \times 10^3$.